Face Detection in Unconstrained Environments

Haar Cascade, VGG-16 e YOLO

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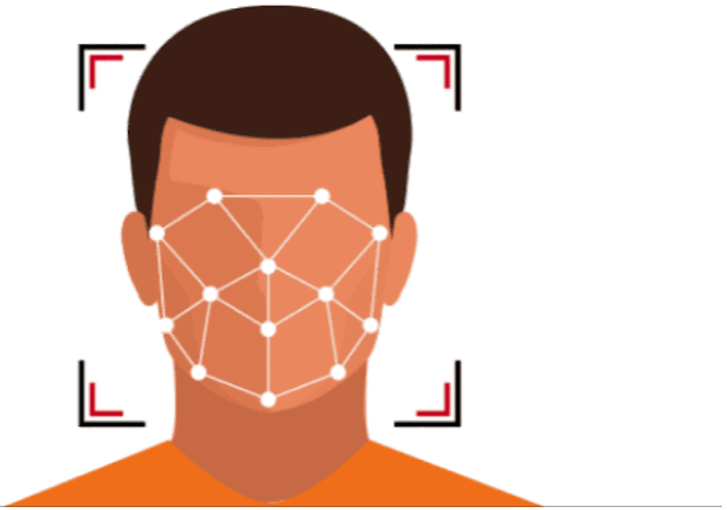

ARTIFICIAL INTELLIGENCE APPLICATIONS | 2026/2027

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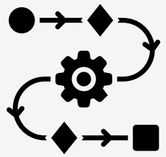
Introduction



Context

- FACIAL DETECTION ALLOWS LOCATING FACES IN IMAGES.
- IT REPRESENTS A CRUCIAL STAGE IN SOFTWARE APPLICATIONS.
- EXAMPLE: BIOMETRIC VERIFICATION

Objectives

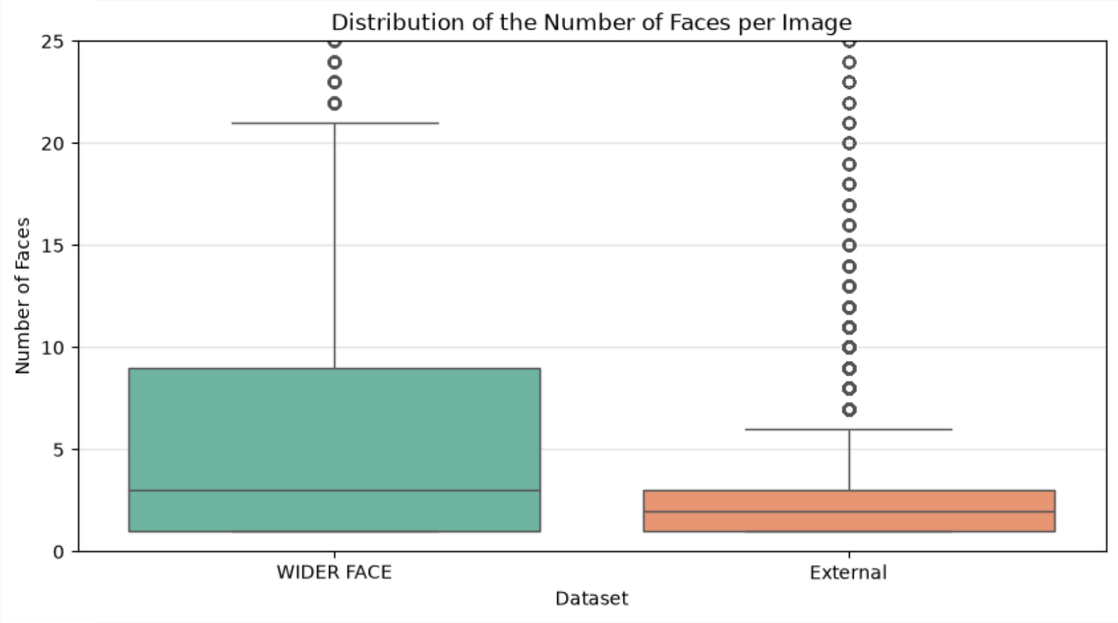
- COMPARE HAAR CASCADE, VGG-16 AND YOLO'S PERFORMANCE ON FACE DETECTION .

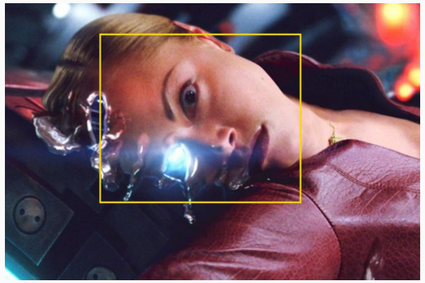
Challenges

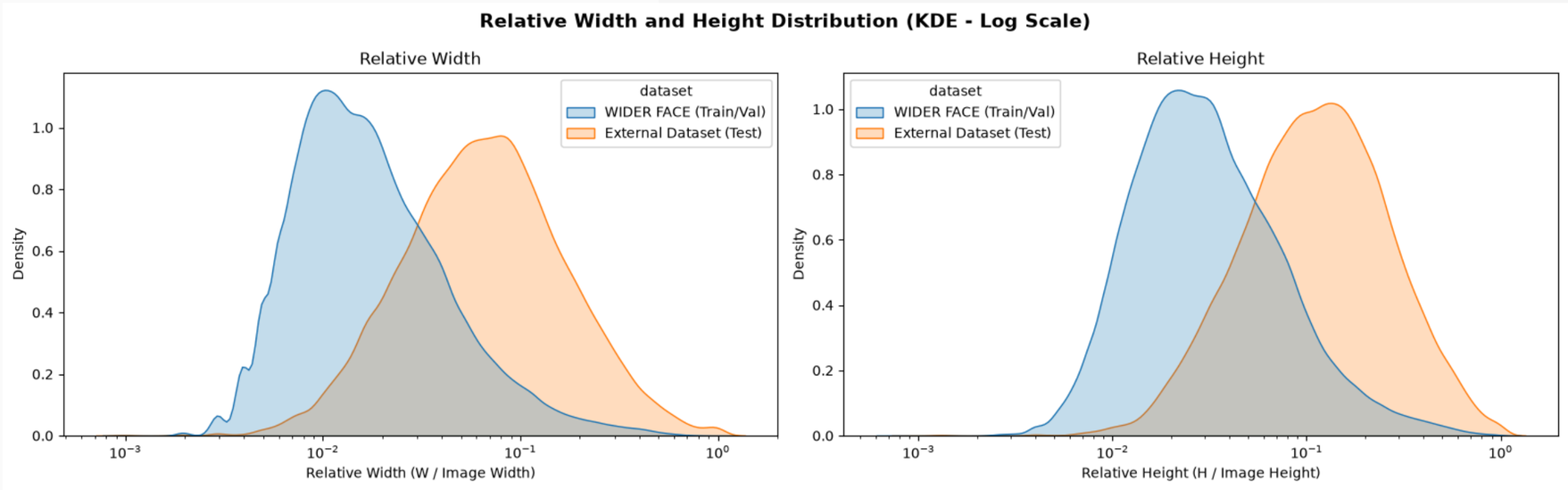
- TRAINING TIME AND COMPUTACIONAL POWER LIMITATIONS.

Methodology: Dataset Analysis

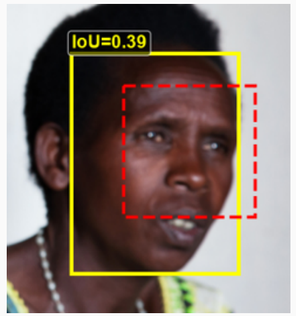
- Main Dataset: WIDER FACE (train and validation)
- External Dataset: Face-Detection-Dataset from Kaggle (test)

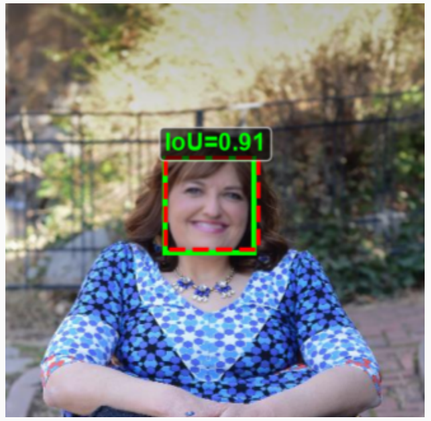




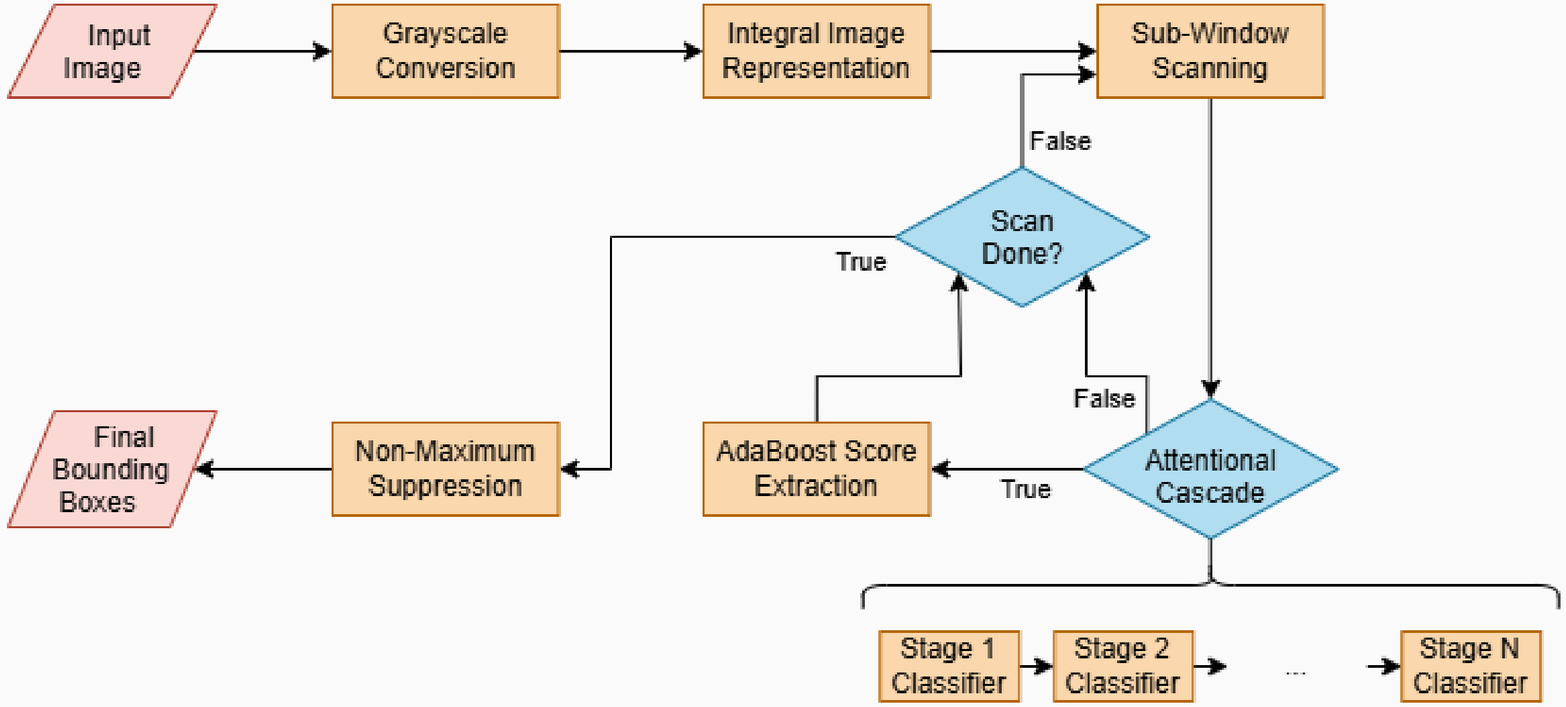


Methodology: Haar Cascade

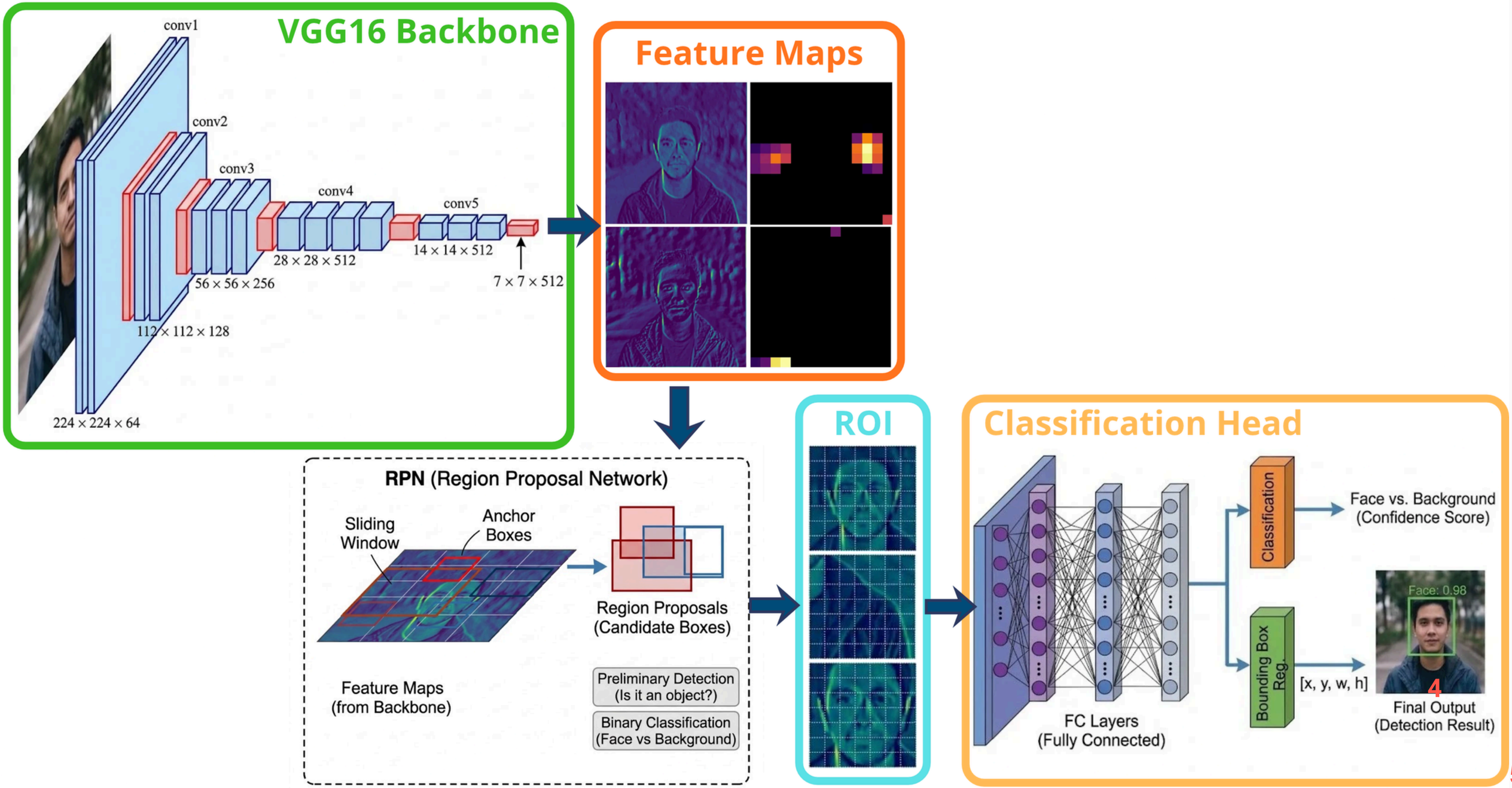




- Base Model
- Hyperparameter Optimization
  - *scaleFactor*
  - *minNeighbors*
  - *minSize*
- Confidence Extraction: *detectMultiScale3*
- Spatial Sensitivity Analysis: Intersection over Union (IoU)



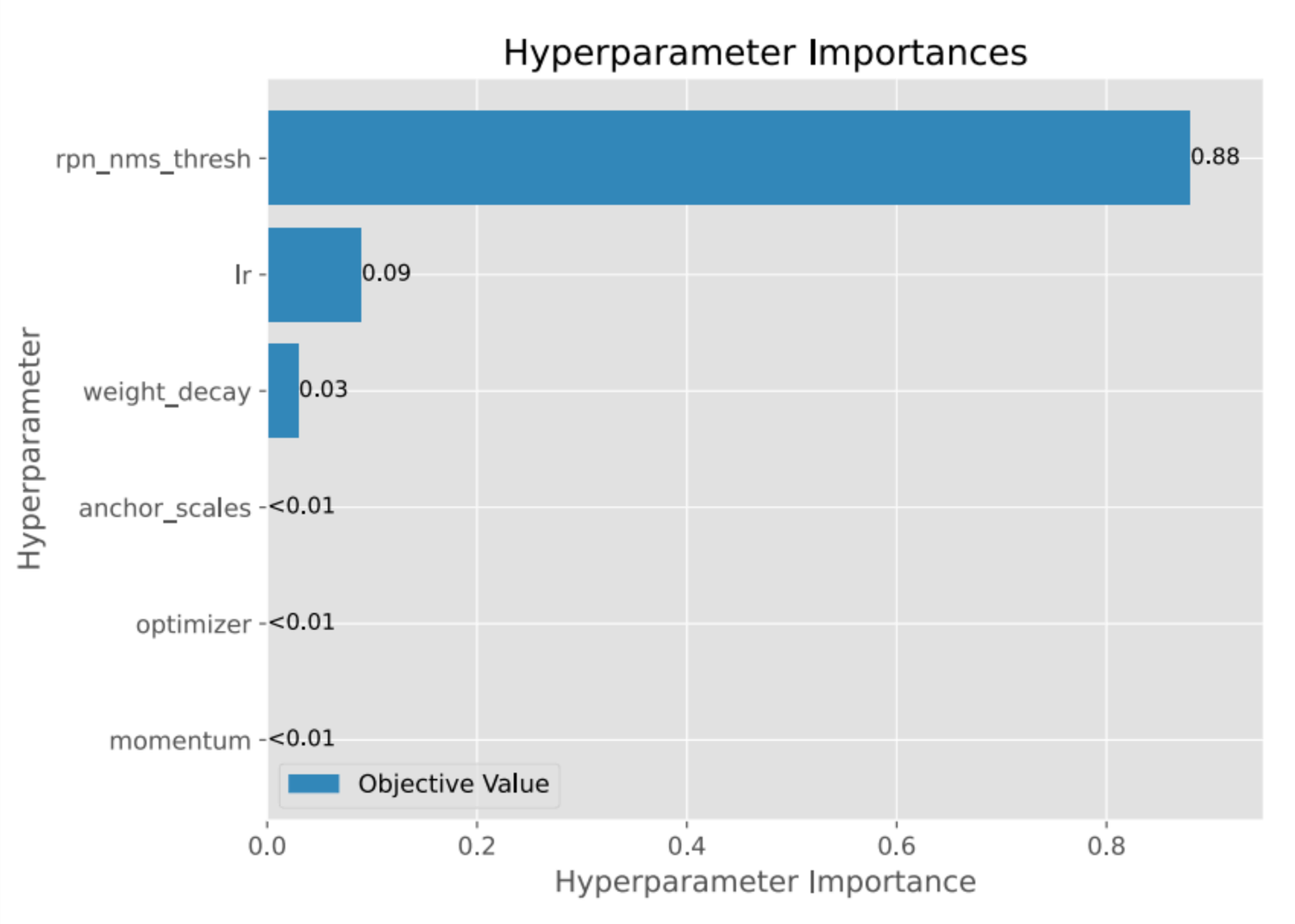
Faster R-CNN with VGG-16



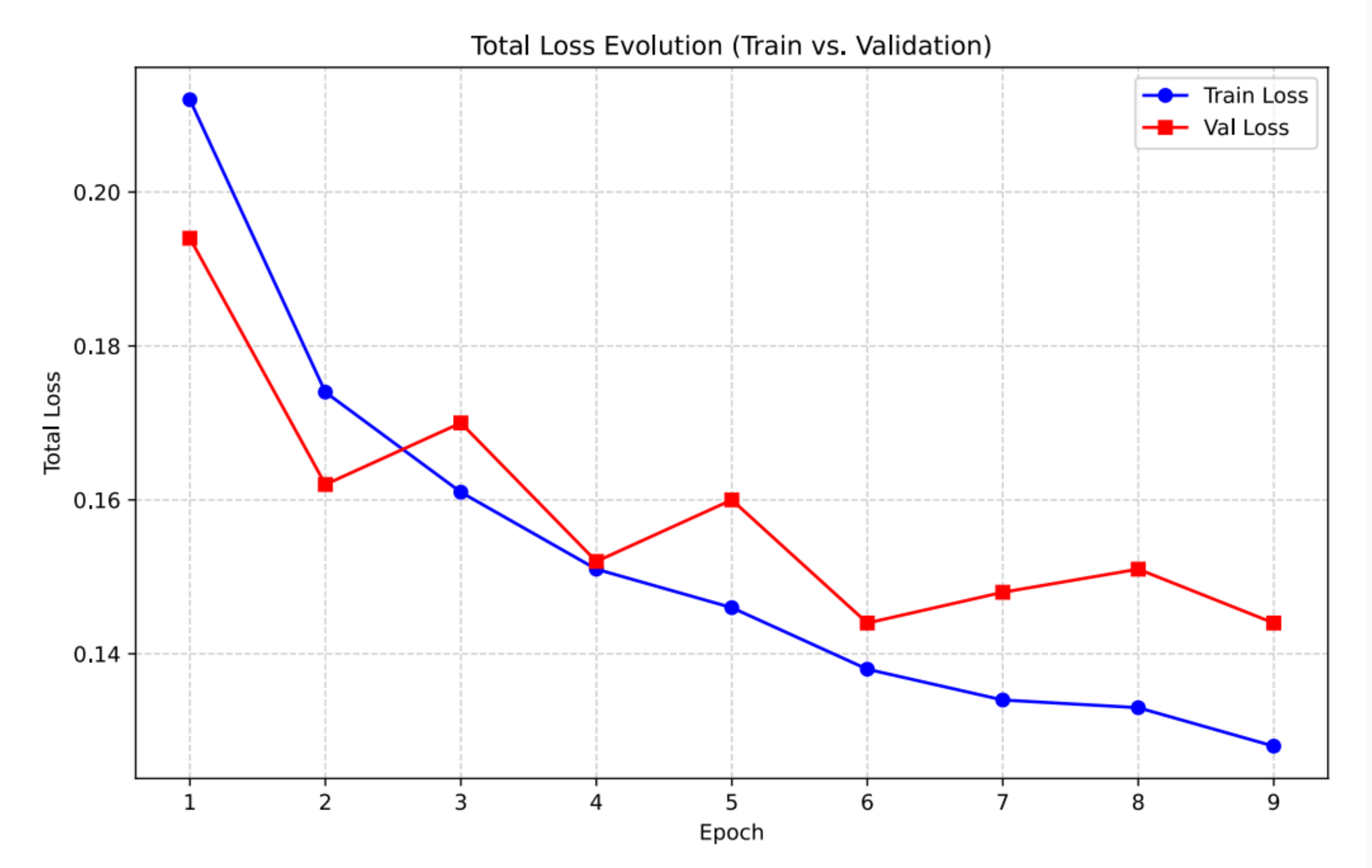
Hyperparameter Tuning Faster R-CNN

OPTUNA

- *25 trials with 5 epochs each:*
  - *20 completed*
  - *5 pruned*
- *Best configuration found:*
  - *rpn_nms_thresh: 0.50*
  - *Learning Rate (lr):* 2.76×10^{-5}
  - *Optimizer: Adam*
  - *Weight Decay:* 9.94×10^{-5}
  - *Anchor Scales: Medium*



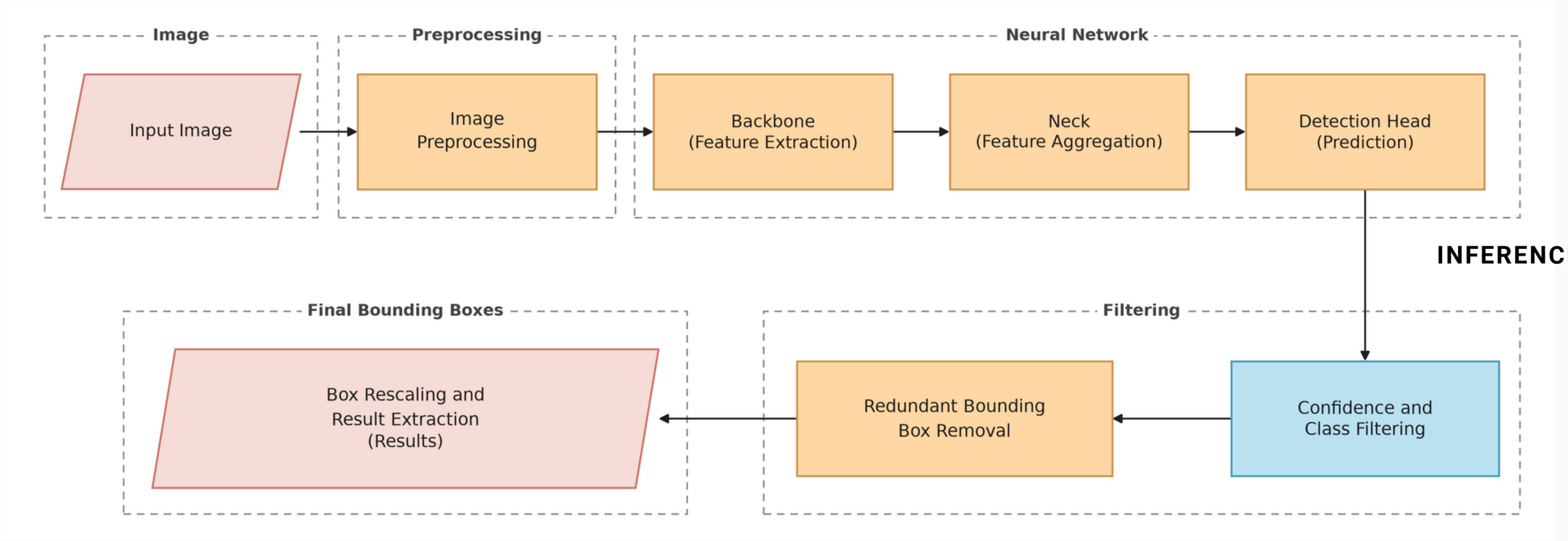
Faster R-CNN Training



Methodology: YOLO

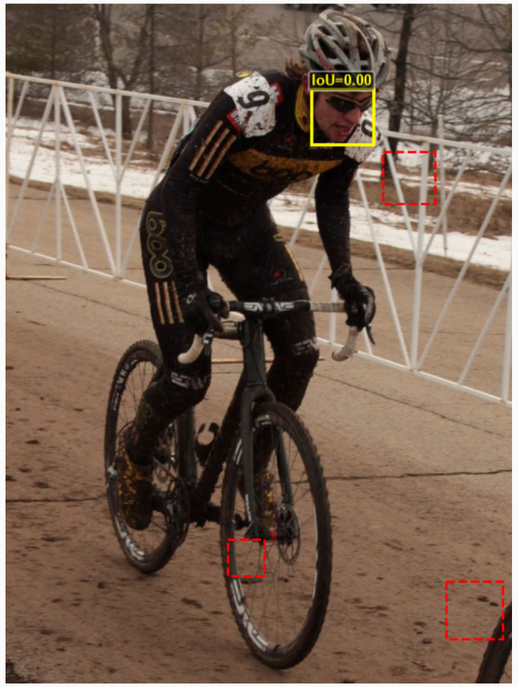
YOU ONLY LOOK ONCE

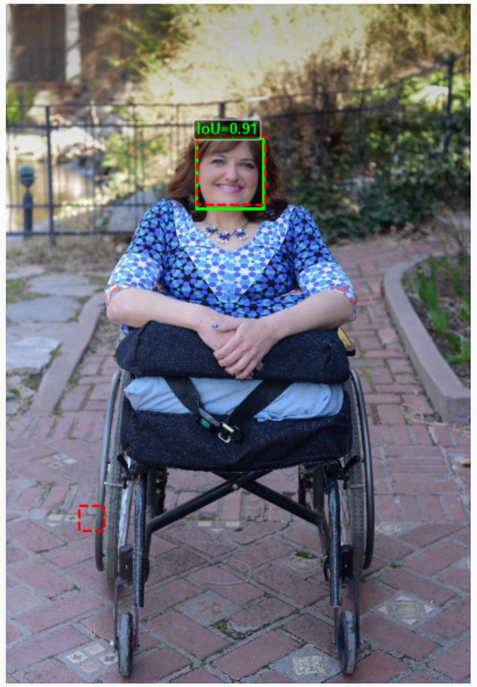
- Models: **YOLO26n base**, **YOLO26n fine-tuned**, **YOLOv8n-Face** and **YOLO26n-Face**.
- Fine-tuning: YOLO26n on WIDER FACE.
- Training setup: **100 epochs**, **640 × 640 images**, **batch size 8** and **NVIDIA Tesla T4 GPU** on Google Colab.
- Evaluation: 16,733 Kaggle images (Dataset), measuring **detection quality** and **inference time**.

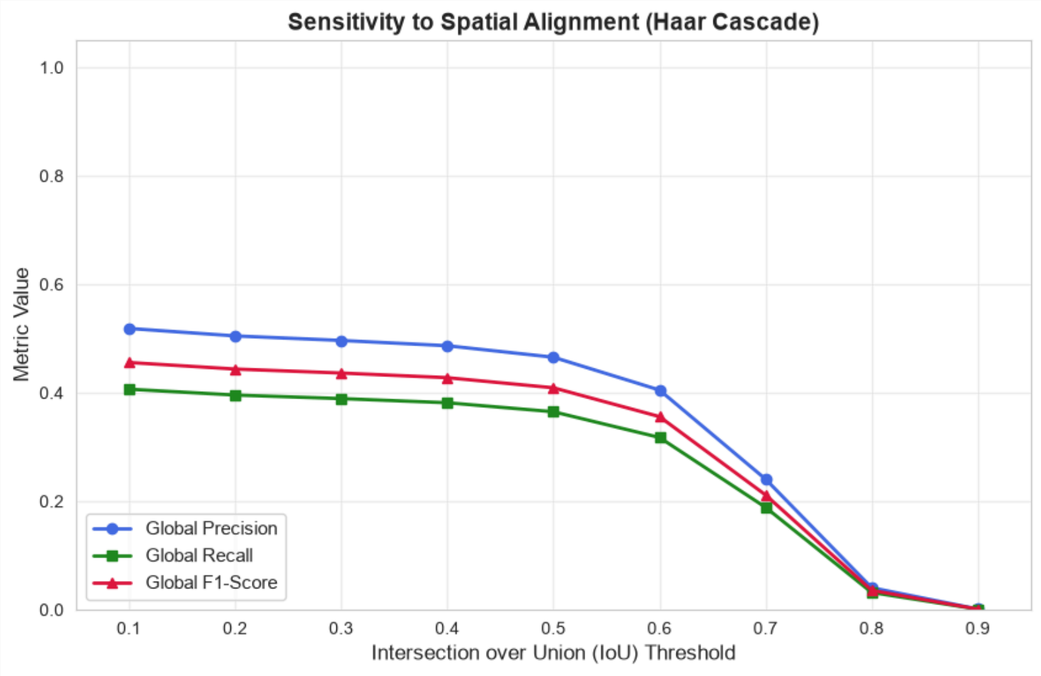


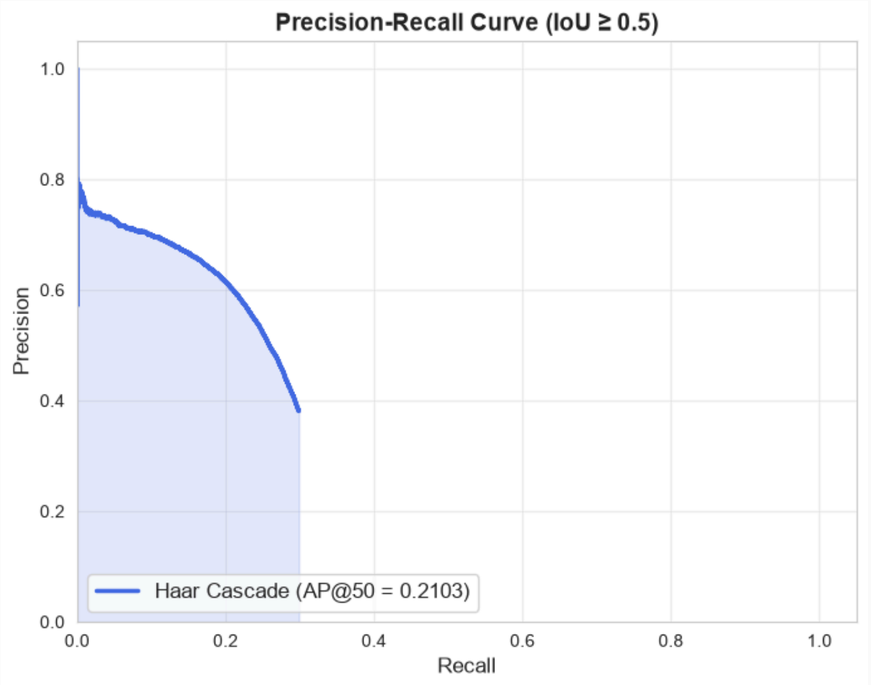
Results: Haar Cascade

- **General Performance:** Average Precision (AP@0.50) = 21.03%
- **Sctructural Limitations:** PR curve stops at 0.3 of Recall
- **IoU Sensitivity:** Abrupt metric loss for IoU ≥ 0.6
- **Inference Time:** Average inference time of 107.82 ms per image.

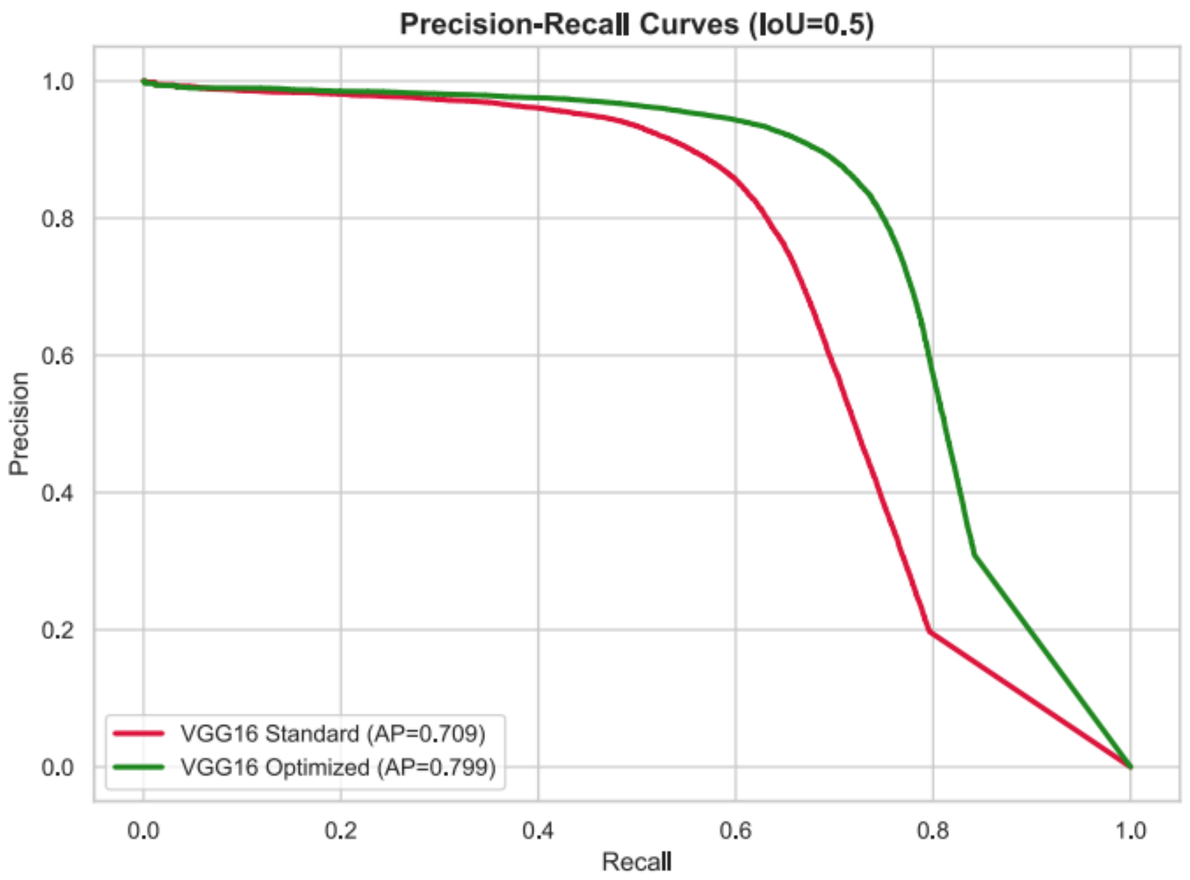


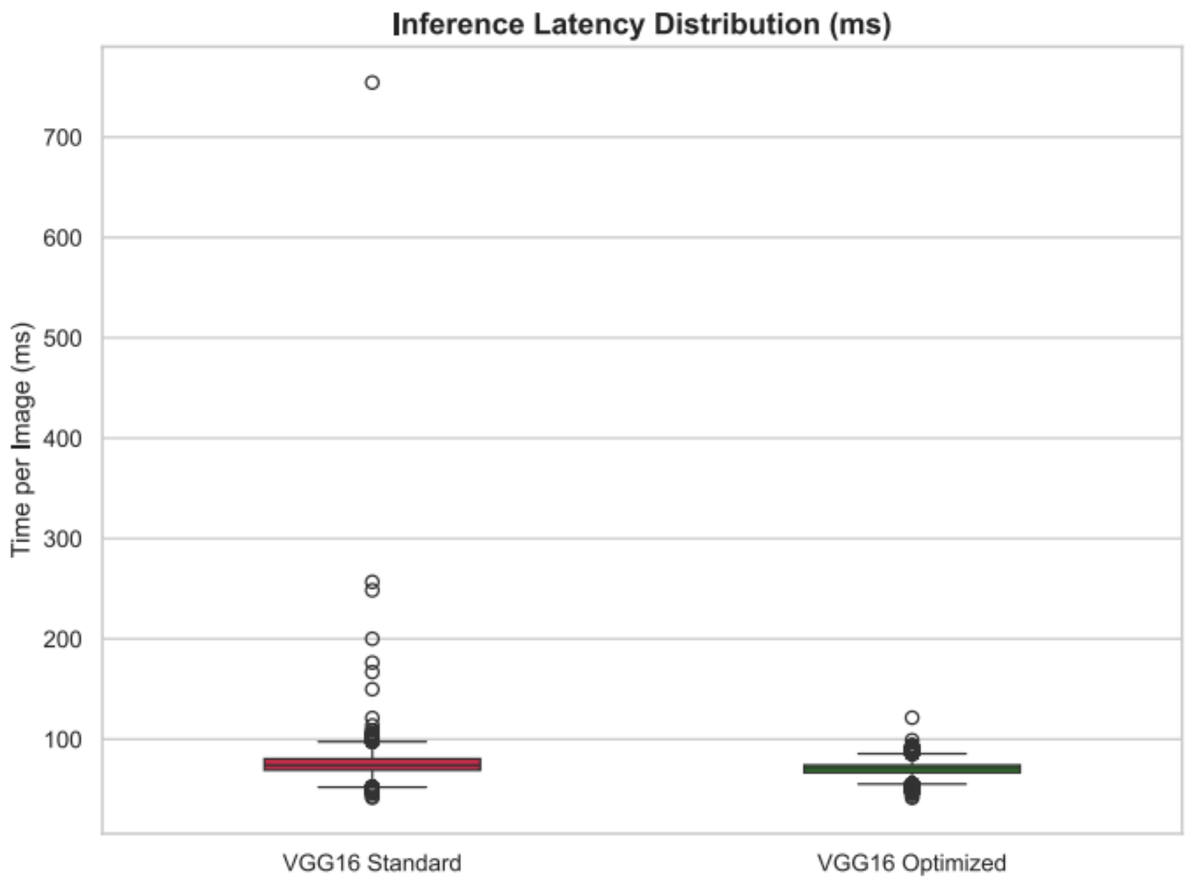






Results: VGG-16



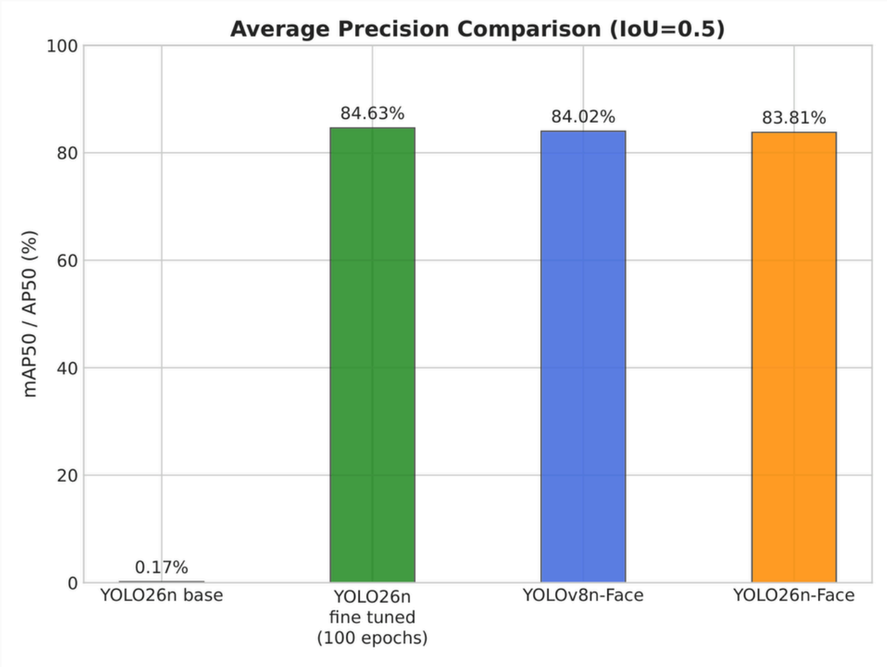


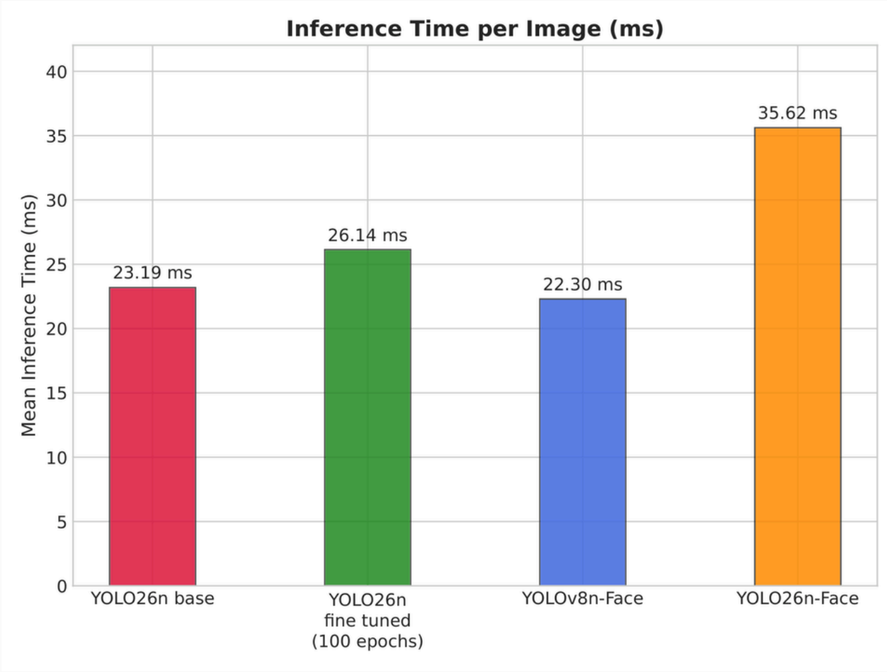
Results: YOLO

- **Highest AP:**
  Fine-tuned YOLO26n

- **Lowest Inference Time per image:**
  YOLOv8n-Face

- **Trade-off:**
  Fine-tuned model achieved slightly higher AP at an additional 3.84 ms per image.





Conclusion

- **Haar Cascade:** limited performance in unconstrained scenes, achieving 21.03% AP.
- **VGG-16:** Optuna optimisation improved AP from 70.90% to 79.90%.
- **YOLO:** achieved strongest balance between detection quality and inference speed.
- **Best Model:** fine-tuned YOLO26n for the highest AP, or YOLOv8n-Face for the lowest mean latency.

Model	Hardware	AP@0.50 (%)	Mean Inference Time (ms)
Viola-Jones (Haar Cascade)	CPU	21.03	107.82
VGG-16 (Standard)	GPU	70.90	74.18
VGG-16 (Optimized)	GPU	79.90	69.85
YOLO26n (Base)	GPU	0.17	23.19
YOLOv8n-Face	GPU	84.02	22.30
YOLO26n (Fine-tuned)	GPU	84.63	26.14
YOLO26n-Face	GPU	83.81	35.62

DEMO

Q&A